



American International University-Bangladesh (AIUB)

Department of Computer Science and Engineering (CSE)

Faculty of Science & Technology (FST)

Spring 2022-23

Course: Introduction to Data Science

Course Code: CSC 4180

Section: E

Final Term Project

Submitted by:

Fahim Mahmud Bhuiyan

ID: 20-42970-1

Department: CSE

Submitted To:

Tohedul Islam

Assistant Professor

Department of Computer Science

Submitted Date: 29th April 2023

Applying the K-means clustering algorithm to Country Data Dataset

Introduction: K-Means Clustering is a technique for unsupervised machine learning that divides the unlabelled dataset into several clusters. K-means clustering is a method of vector quantization technique that originated in signal processing, is to divide n observations into k clusters, each of which has a prototype (cluster centroid or cluster center) that each observation belongs to. In this project, there were trying to apply the K-means clustering algorithm to Country Data Dataset using R language.

“Country data” dataset was found from Kaggle.

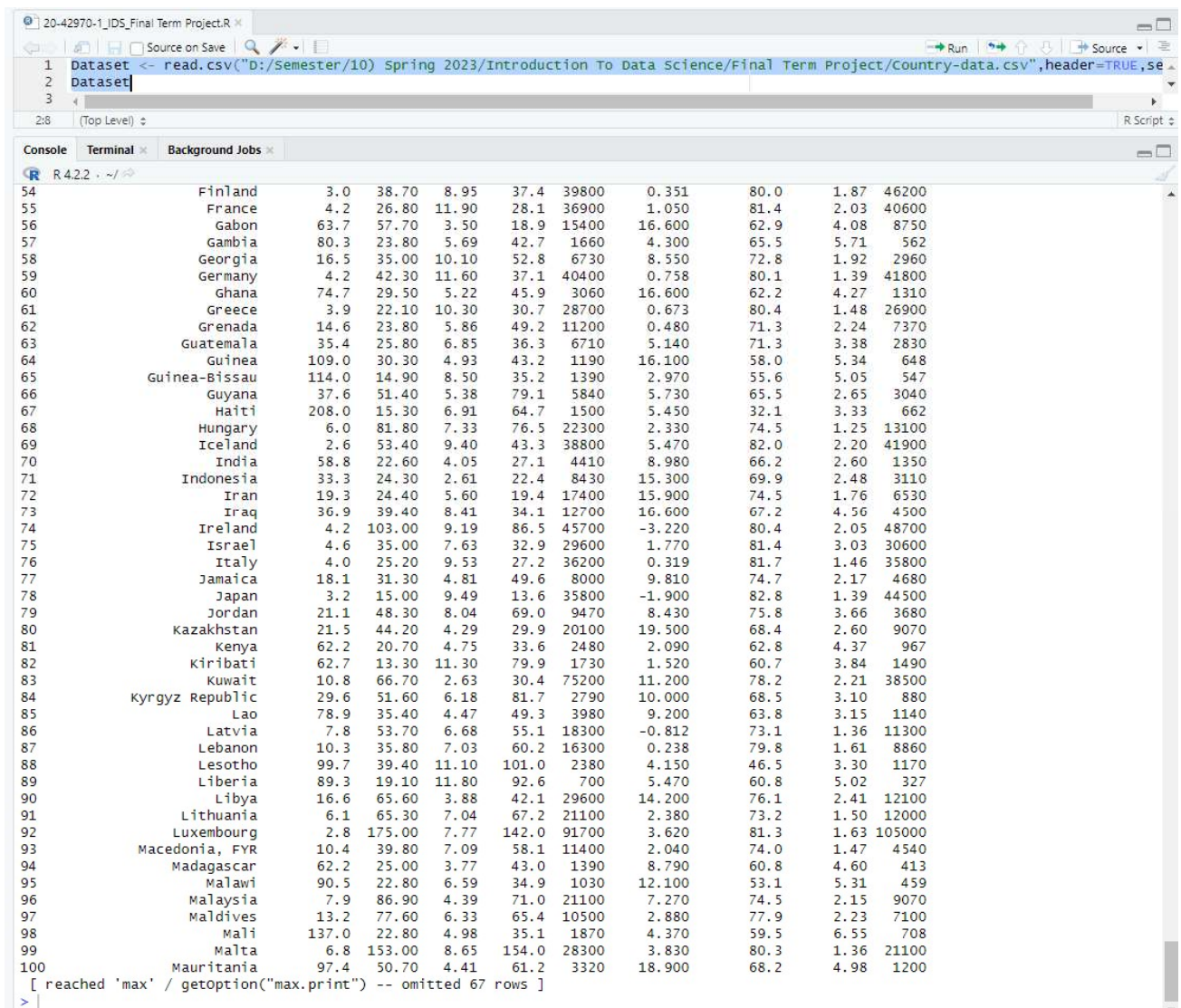
(<https://www.kaggle.com/datasets/rohan0301/unsupervised-learning-on-country-data>).

Project's steps and the code:

1) Import and Print the Data Set

```
Dataset <- read.csv("D:/Semester/10) Spring 2023/Introduction To Data Science/Final Term Project/Country-data.csv",header=TRUE,sep=",")
```

Dataset



The screenshot shows an R Studio interface with a script editor and a console. The script editor contains the following code:

```
1 Dataset <- read.csv("D:/Semester/10) Spring 2023/Introduction To Data Science/Final Term Project/Country-data.csv",header=TRUE,se  
2 Dataset  
3
```

The console displays the output of the code, showing the first 100 rows of the dataset. The data is a table with 11 columns and 100 rows. The columns are: Country, GDP, Life Expectancy, Literacy Rate, Fertility Rate, Infant Mortality Rate, Unemployment Rate, Inflation Rate, Exchange Rate, and two unnamed columns. The data is sorted by GDP in descending order.

Country	GDP	Life Expectancy	Literacy Rate	Fertility Rate	Infant Mortality Rate	Unemployment Rate	Inflation Rate	Exchange Rate		
Finland	3.0	38.70	8.95	37.4	39800	0.351	80.0	1.87	46200	
France	4.2	26.80	11.90	28.1	36900	1.050	81.4	2.03	40600	
Gabon	63.7	57.70	3.50	18.9	15400	16.600	62.9	4.08	8750	
Gambia	80.3	23.80	5.69	42.7	1660	4.300	65.5	5.71	562	
Georgia	16.5	35.00	10.10	52.8	6730	8.550	72.8	1.92	2960	
Germany	4.2	42.30	11.60	37.1	40400	0.758	80.1	1.39	41800	
Ghana	74.7	29.50	5.22	45.9	3060	16.600	62.2	4.27	1310	
Greece	3.9	22.10	10.30	30.7	28700	0.673	80.4	1.48	26900	
Grenada	14.6	23.80	5.86	49.2	11200	0.480	71.3	2.24	7370	
Guatemala	35.4	25.80	6.85	36.3	6710	5.140	71.3	3.38	2830	
Guinea	109.0	30.30	4.93	43.2	1190	16.100	58.0	5.34	648	
Guinea-Bissau	114.0	14.90	8.50	35.2	1390	2.970	55.6	5.05	547	
Guyana	37.6	51.40	5.38	79.1	5840	5.730	65.5	2.65	3040	
Haiti	208.0	15.30	6.91	64.7	1500	5.450	32.1	3.33	662	
Hungary	6.0	81.80	7.33	76.5	22300	2.330	74.5	1.25	13100	
Iceland	2.6	53.40	9.40	43.3	38800	5.470	82.0	2.20	41900	
India	58.8	22.60	4.05	27.1	4410	8.980	66.2	2.60	1350	
Indonesia	33.3	24.30	2.61	22.4	8430	15.300	69.9	2.48	3110	
Iran	19.3	24.40	5.60	19.4	17400	15.900	74.5	1.76	6530	
Iraq	36.9	39.40	8.41	34.1	12700	16.600	67.2	4.56	4500	
Ireland	4.2	103.00	9.19	86.5	45700	-3.220	80.4	2.05	48700	
Israel	4.6	35.00	7.63	32.9	29600	1.770	81.4	3.03	30600	
Italy	4.0	25.20	9.53	27.2	36200	0.319	81.7	1.46	35800	
Jamaica	18.1	31.30	4.81	49.6	8000	9.810	74.7	2.17	4680	
Japan	3.2	15.00	9.49	13.6	35800	-1.900	82.8	1.39	44500	
Jordan	21.1	48.30	8.04	69.0	9470	8.430	75.8	3.66	3680	
Kazakhstan	21.5	44.20	4.29	29.9	20100	19.500	68.4	2.60	9070	
Kenya	62.2	20.70	4.75	33.6	2480	2.090	62.8	4.37	967	
Kiribati	62.7	13.30	11.30	79.9	1730	1.520	60.7	3.84	1490	
Kuwait	10.8	66.70	2.63	30.4	75200	11.200	78.2	2.21	38500	
Kyrgyz Republic	29.6	51.60	6.18	81.7	2790	10.000	68.5	3.10	880	
Lao	78.9	35.40	4.47	49.3	3980	9.200	63.8	3.15	1140	
Latvia	7.8	53.70	6.68	55.1	18300	-0.812	73.1	1.36	11300	
Lebanon	10.3	35.80	7.03	60.2	16300	0.238	79.8	1.61	8860	
Lesotho	99.7	39.40	11.10	101.0	2380	4.150	46.5	3.30	1170	
Madagascar	89.3	19.10	11.80	92.6	700	5.470	60.8	5.02	327	
Malawi	16.6	65.60	3.88	42.1	29600	14.200	76.1	2.41	12100	
Lithuania	6.1	65.30	7.04	67.2	21100	2.380	73.2	1.50	12000	
Luxembourg	2.8	175.00	7.77	142.0	91700	3.620	81.3	1.63	105000	
Macedonia, FYR	10.4	39.80	7.09	58.1	11400	2.040	74.0	1.47	4540	
Madagascar	62.2	25.00	3.77	43.0	1390	8.790	60.8	4.60	413	
Malawi	90.5	22.80	6.59	34.9	1030	12.100	53.1	5.31	459	
Malaysia	7.9	86.90	4.39	71.0	21100	7.270	74.5	2.15	9070	
Maldives	13.2	77.60	6.33	65.4	10500	2.880	77.9	2.23	7100	
Mali	137.0	22.80	4.98	35.1	1870	4.370	59.5	6.55	708	
Malta	6.8	153.00	8.65	154.0	28300	3.830	80.3	1.36	21100	
Mauritania	97.4	50.70	4.41	61.2	3320	18.900	68.2	4.98	1200	

[reached 'max' / getOption("max.print") -- omitted 67 rows]

2) Study the Dataset

names(Dataset)

summary(Dataset)

str(Dataset)

sum(is.na(Dataset))

head(Dataset)

```
20-42970-1_IDS_Final Term Project.R*
1 Dataset <- read.csv("D:/Semester/10) Spring 2023/Introduction To Data Science/Final Term Project/Country-data.csv",head
2 Dataset
3 names(Dataset)
4 summary(Dataset)
5 str(Dataset)
6 sum(is.na(Dataset))
7
8 head(Dataset)
9
```

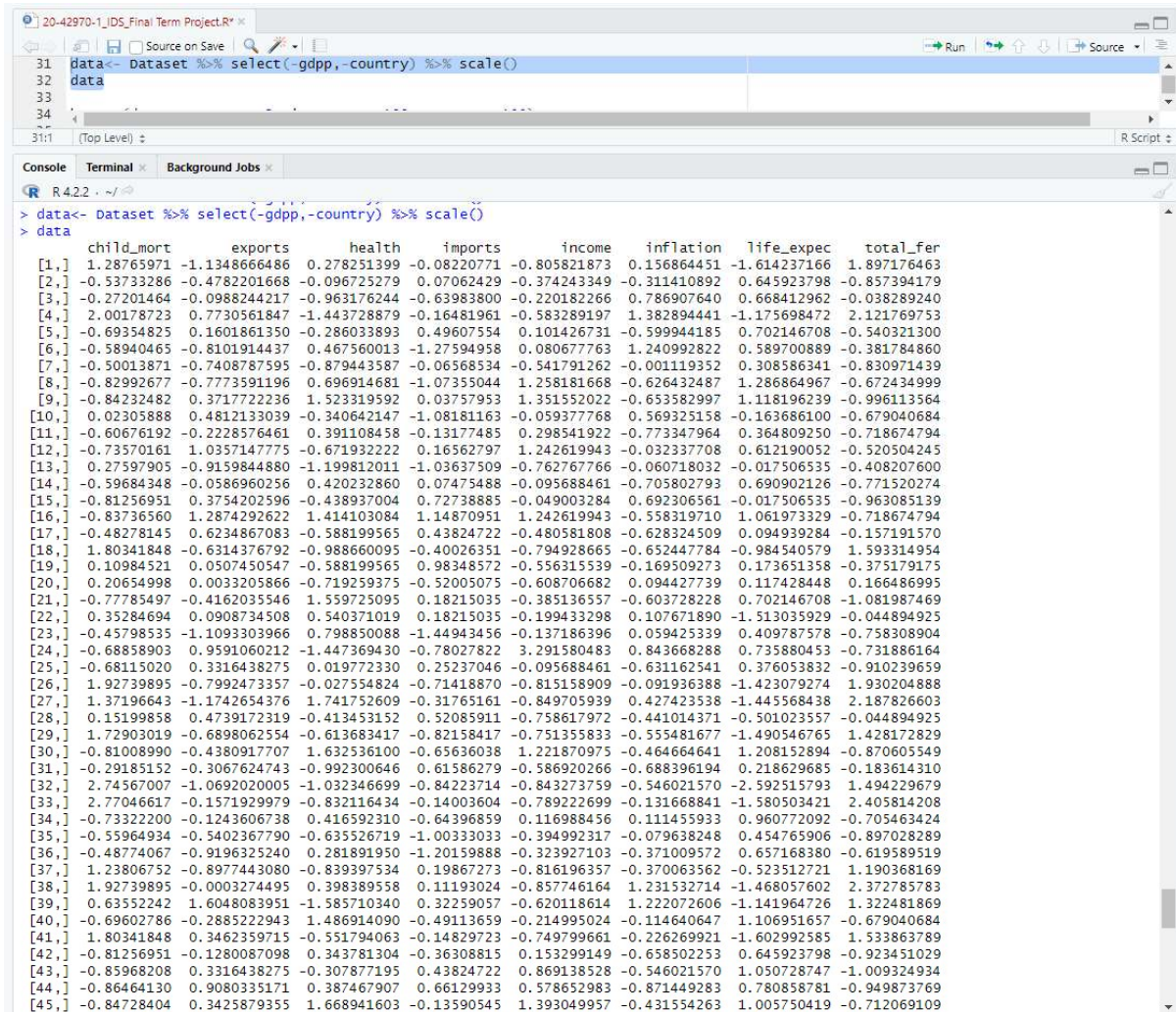
```
R 4.2.2. ~/
[ reached 'max' / getOption("max.print") -- omitted 67 rows ]
> names(Dataset)
[1] "country" "child_mort" "exports" "health" "imports" "income" "inflation" "life_expec" "total_fer"
[10] "gdpp"
> summary(Dataset)
country      child_mort      exports      health      imports      income      inflation
Length:167   Min.   : 2.60   Min.   : 0.109   Min.   : 1.810   Min.   : 0.0659   Min.   : 609   Min.   : -4.210
Class:character 1st Qu.: 8.25   1st Qu.: 23.800   1st Qu.: 4.920   1st Qu.: 30.2000   1st Qu.: 3355   1st Qu.: 1.810
Mode :character Median : 19.30   Median : 35.000   Median : 6.320   Median : 43.3000   Median : 9960   Median : 5.390
               Mean : 38.27   Mean : 41.109   Mean : 6.816   Mean : 46.8902   Mean : 17145   Mean : 7.782
               3rd Qu.: 62.10   3rd Qu.: 51.350   3rd Qu.: 8.600   3rd Qu.: 58.7500   3rd Qu.: 22800   3rd Qu.: 10.750
               Max.   :208.00   Max.   :200.000   Max.   :17.900   Max.   :174.0000   Max.   :125000   Max.   :104.000

life_expec    total_fer    gdpp
Min.   :32.10   Min.   :1.150   Min.   : 231
1st Qu.:65.30   1st Qu.:1.795   1st Qu.: 1330
Median :73.10   Median :2.410   Median : 4660
Mean   :70.56   Mean   :2.948   Mean   :12964
3rd Qu.:76.80   3rd Qu.:3.880   3rd Qu.:14050
Max.   :82.80   Max.   :7.490   Max.   :105000
> str(Dataset)
'data.frame': 167 obs. of 10 variables:
 $ country : chr "Afghanistan" "Albania" "Algeria" "Angola" ...
 $ child_mort: num 90.2 16.6 27.3 119 10.3 14.5 18.1 4.8 4.3 39.2 ...
 $ exports : num 10 28 38.4 62.3 45.5 18.9 20.8 19.8 51.3 54.3 ...
 $ health : num 7.58 6.55 4.17 2.85 6.03 8.1 4.4 8.73 11 5.88 ...
 $ imports : num 44.9 48.6 31.4 42.9 58.9 16 45.3 20.9 47.8 20.7 ...
 $ income : int 1610 9930 12900 5900 19100 18700 6700 41400 43200 16000 ...
 $ inflation: num 9.44 4.49 16.1 22.4 1.44 20.9 7.77 1.16 0.873 13.8 ...
 $ life_expec: num 56.2 76.3 76.5 60.1 76.8 75.8 73.3 82 80.5 69.1 ...
 $ total_fer : num 5.82 1.65 2.89 6.16 2.13 2.37 1.69 1.93 1.44 1.92 ...
 $ gdpp : int 553 4090 4460 3530 12200 10300 3220 51900 46900 5840 ...
> sum(is.na(Dataset))
[1] 0
>
> head(Dataset)
  country child_mort exports health imports income inflation life_expec total_fer gdpp
1 Afghanistan  90.2   10.0   7.58   44.9   1610    9.44    56.2    5.82   553
2 Albania     16.6   28.0   6.55   48.6   9930    4.49    76.3    1.65  4090
3 Algeria     27.3   38.4   4.17   31.4  12900   16.10    76.5    2.89  4460
4 Angola     119.0   62.3   2.85   42.9   5900   22.40    60.1    6.16  3530
5 Antigua and Barbuda 10.3   45.5   6.03   58.9  19100    1.44    76.8    2.13 12200
6 Argentina   14.5   18.9   8.10   16.0  18700   20.90    75.8    2.37 10300
```


3) Create the Standardize Continuous Variables

```
data<- Dataset %>% select(-gdp,-country) %>% scale()
```

data



The screenshot shows the RStudio interface. The script editor contains the command `data<- Dataset %>% select(-gdp,-country) %>% scale()` on line 31. The console shows the execution of this command, resulting in a data frame with 45 rows and 10 columns. The columns are: `child_mort`, `exports`, `health`, `imports`, `income`, `inflation`, `life_expec`, and `total_fer`. The data is displayed in a grid format, with the first few rows showing values for these variables across 45 different countries or regions.

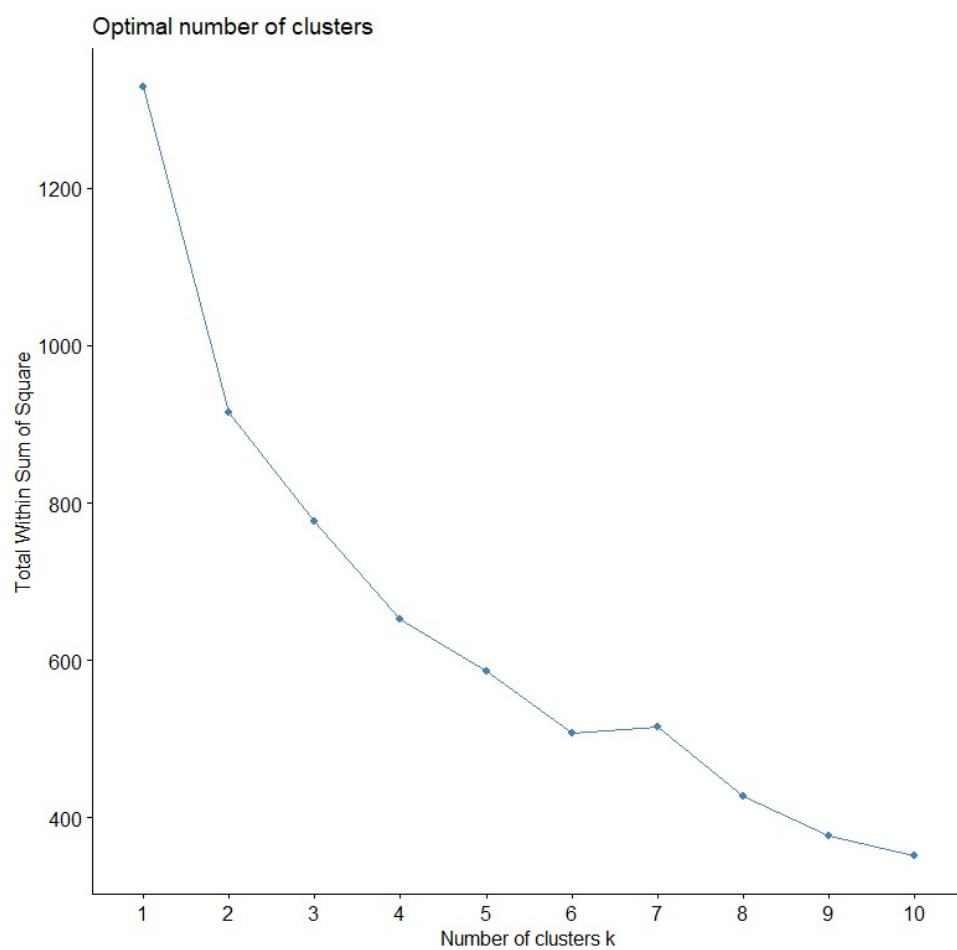
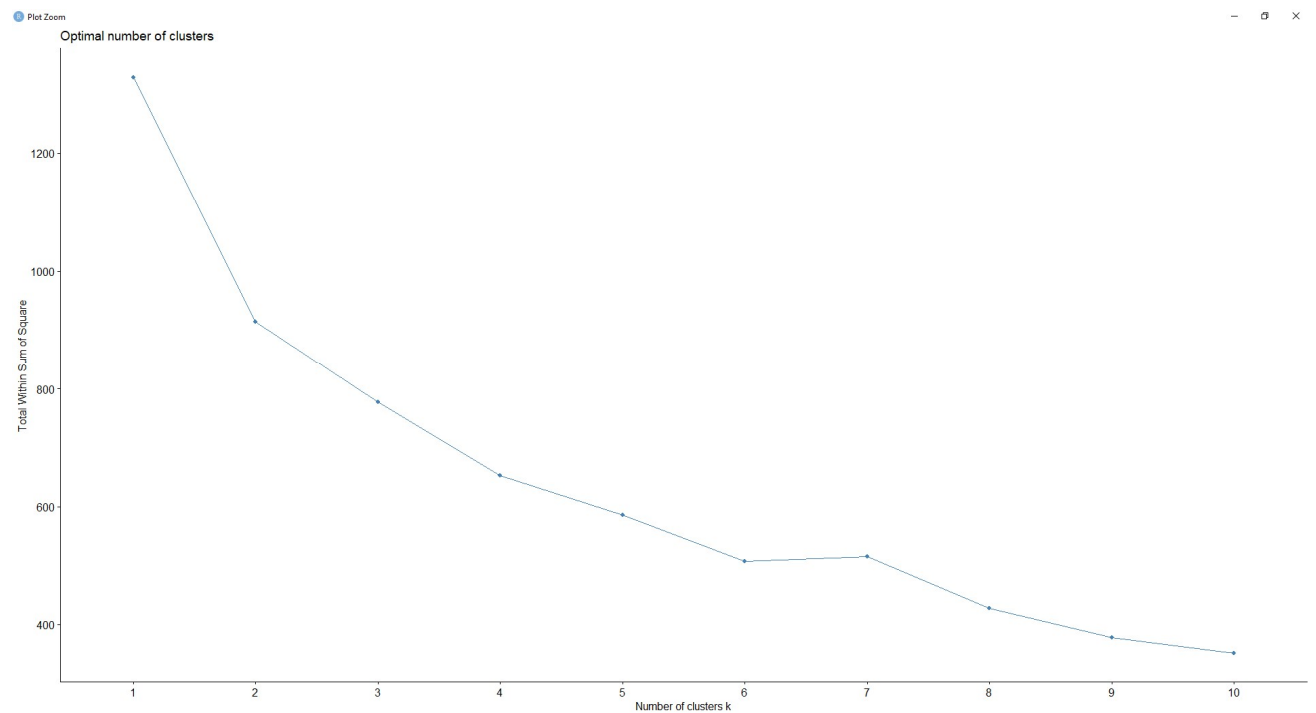
```
> data<- Dataset %>% select(-gdp,-country) %>% scale()
> data
```

	child_mort	exports	health	imports	income	inflation	life_expec	total_fer
[1,]	1.28765971	-1.1348666486	0.278251399	-0.08220771	-0.805821873	0.156864451	-1.614237166	1.897176463
[2,]	-0.53733286	-0.4782201668	-0.096725279	0.07062429	-0.374243349	-0.311410892	0.645923798	-0.857394179
[3,]	-0.27201464	-0.0988244217	-0.963176244	-0.63983800	-0.220182266	0.786907640	0.668412962	-0.038289240
[4,]	2.00178723	0.7730561847	-1.443728879	-0.16481961	-0.583289197	1.382894441	-1.175698472	2.121769753
[5,]	-0.69354825	0.1601861350	-0.286033893	0.49607554	0.101426731	-0.599944185	0.702146708	-0.540321300
[6,]	-0.58940465	-0.8101914437	0.467560013	-1.27594958	0.080677763	1.240992822	0.589700889	-0.381784860
[7,]	-0.50013871	-0.7408787595	-0.879443587	-0.06568534	-0.541791262	-0.001119352	0.308586341	-0.830971439
[8,]	-0.82992677	-0.7773591196	0.696914681	-1.07355044	1.258181668	-0.626432487	1.286864967	-0.672434999
[9,]	-0.84232482	0.371772236	1.523319592	0.03757953	1.351552022	-0.653582997	1.118196239	-0.996113564
[10,]	0.02305888	0.4812133039	-0.340642147	-1.08181163	-0.059377768	0.569325158	-0.163686100	-0.679040684
[11,]	-0.60876192	-0.2228576461	0.391108458	-0.13177485	0.298541922	-0.773347964	0.364809250	-0.718674794
[12,]	-0.73570161	-0.0357147775	-0.671932222	0.16562797	1.242619943	-0.032337708	0.612190052	-0.520504245
[13,]	0.27597905	-0.9159844880	-1.199812011	-1.03637509	-0.762767766	-0.060718032	-0.017506535	-0.408207600
[14,]	-0.59684348	-0.0586960256	0.420232860	0.07475488	-0.095688461	-0.705802793	0.690902126	-0.771520274
[15,]	-0.81256951	0.3754202596	-0.438937004	0.72738885	-0.049003284	0.692306561	-0.017506535	-0.963085139
[16,]	-0.83736560	1.2874292622	1.414103084	1.14870951	1.242619943	-0.558319710	1.061973329	-0.718674794
[17,]	-0.48278145	0.6234867083	-0.588199565	0.43824722	-0.480581808	-0.628324509	0.094939284	-0.157191570
[18,]	1.80341848	-0.6314376792	-0.988660095	-0.40026351	-0.794928665	-0.652447784	-0.984540579	1.593314954
[19,]	0.10984521	0.0507450547	-0.588199565	0.98348572	-0.556315539	-0.169509273	0.173651358	-0.375179175
[20,]	0.20654998	0.0033205866	-0.719259375	-0.52005075	-0.608706682	0.094427739	0.117428448	0.166486995
[21,]	-0.77785497	-0.4162035546	1.559725095	0.18215035	-0.385136557	-0.603728228	0.702146708	-1.081987469
[22,]	0.35284694	0.0908734508	0.540371019	0.18215035	-0.199433298	0.107671890	-1.513035929	-0.044894925
[23,]	-0.45798535	-1.1093303966	0.798850088	-1.44943456	-0.137186396	0.059425339	0.409787578	-0.758308904
[24,]	-0.68858903	0.9591060212	-1.447369430	-0.78027822	3.291580483	0.843668288	0.735880453	-0.731886164
[25,]	-0.68115020	0.3316438275	0.019772330	0.25237046	-0.095688461	-0.631162541	0.376053832	-0.910239659
[26,]	1.92739895	-0.7992473357	-0.027554824	-0.71418870	-0.815158909	-0.091936388	-1.423079274	1.930204888
[27,]	1.37196643	-1.1742654376	1.741752609	-0.31765161	-0.849705939	0.427423538	-1.445568438	2.187826603
[28,]	0.15199858	0.4739172319	-0.413453152	0.52085911	-0.758617972	-0.441014371	-0.501023557	-0.044894925
[29,]	1.72903019	-0.6898062554	-0.613683417	-0.82158417	-0.751355833	-0.555481677	-1.490546765	1.428172829
[30,]	-0.81008990	-0.4380917707	1.632536100	-0.65636038	1.221870975	-0.464664641	1.208152894	-0.870605549
[31,]	-0.29185152	-0.3067624743	-0.992300646	0.61586279	-0.586920266	-0.688396194	0.218629685	-0.183614310
[32,]	2.74567007	-1.0692020005	-1.032346699	-0.84223714	-0.843273759	-0.546021570	-2.592515793	1.494229679
[33,]	-2.77046617	-0.1571929979	-0.832116434	-0.14003604	-0.789222699	-0.131668841	-1.580503421	2.405814208
[34,]	-0.73322200	-0.1243606738	0.416592310	-0.64396859	0.116988456	0.111455933	0.960772092	-0.705463424
[35,]	-0.55964934	-0.5402367790	-0.635526719	-1.00333033	-0.394992317	-0.079638248	0.454765906	-0.897028289
[36,]	-0.48774067	-0.9196325240	0.281891950	-1.20159888	-0.323927103	-0.371009572	0.657168380	-0.619589519
[37,]	1.23806752	-0.8977443080	-0.839397534	0.19867273	-0.816196357	-0.370063562	-0.523512721	1.190368169
[38,]	1.92739895	-0.0003274495	0.398389558	0.11193024	-0.857746164	1.231532714	-1.468057602	2.372785783
[39,]	0.63552242	1.6048083951	-1.585710340	0.32259057	-0.620118614	1.222072606	-1.141964726	1.322481869
[40,]	-0.69602786	-0.2885222943	1.486914090	-0.49113659	-0.214995024	-0.114640647	1.106951657	-0.679040684
[41,]	1.80341848	0.3462359715	-0.551794063	-0.14829723	-0.749799661	-0.226269921	-1.602992585	1.533863789
[42,]	-0.81256951	-0.1280087098	0.343781304	-0.36308815	0.153299149	-0.658502253	0.645923798	-0.923451029
[43,]	-0.85968208	0.3316438275	-0.307877195	0.43824722	0.869138528	-0.546021570	1.050728747	-1.009324934
[44,]	-0.86464130	0.9080335171	0.387467907	0.66129933	0.578652983	-0.871449283	0.780858781	-0.949873769
[45,]	-0.84728404	0.3425879355	1.668941603	-0.13590545	1.393049957	-0.431554263	1.005750419	-0.712069109

4) Identify and display the optimal number of Clusters

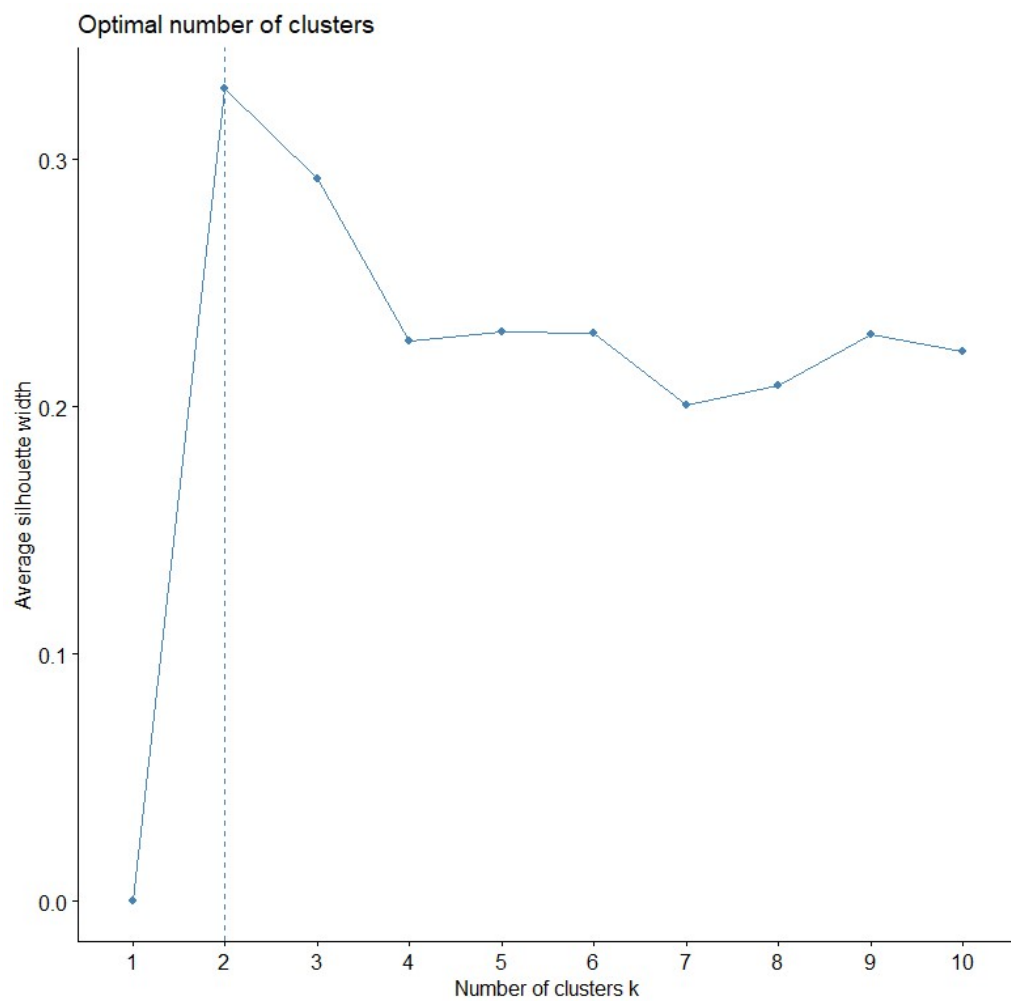
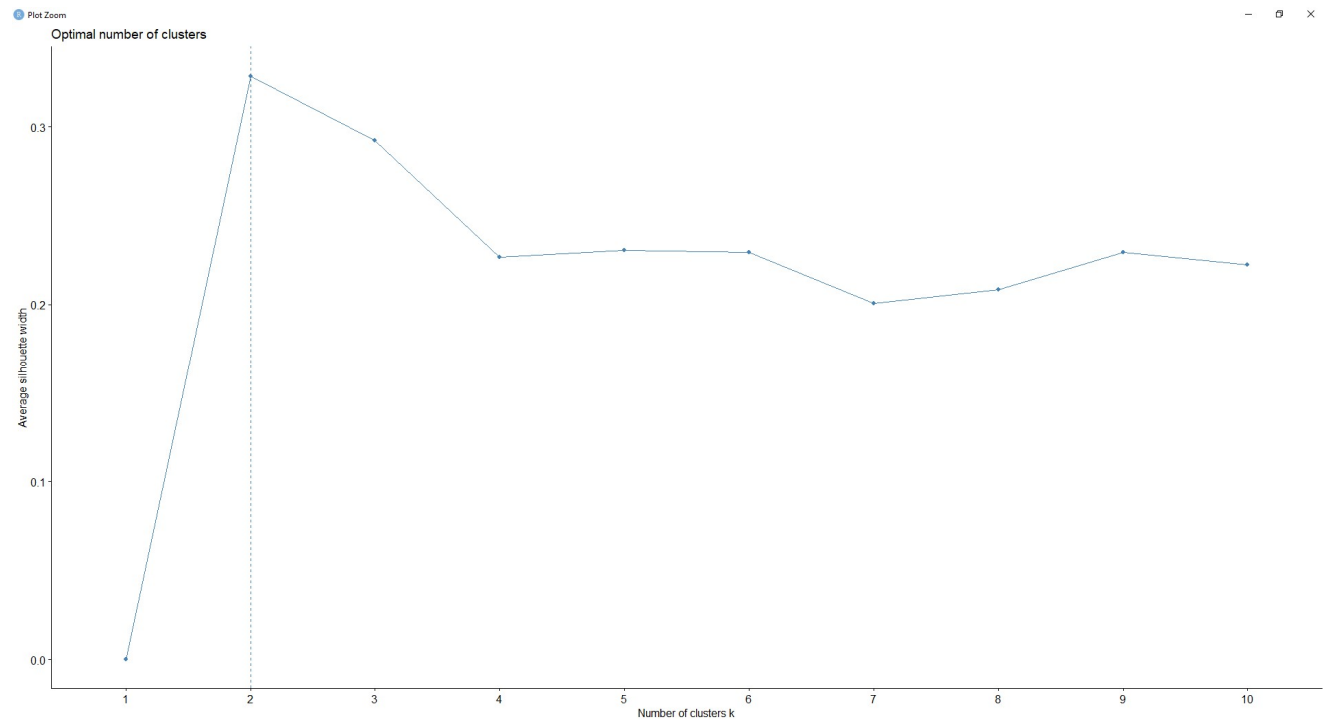
Using WSS Elbow Method:

```
fviz_nbclust(data, kmeans, method = "wss")
```



Using Silhouette Method:

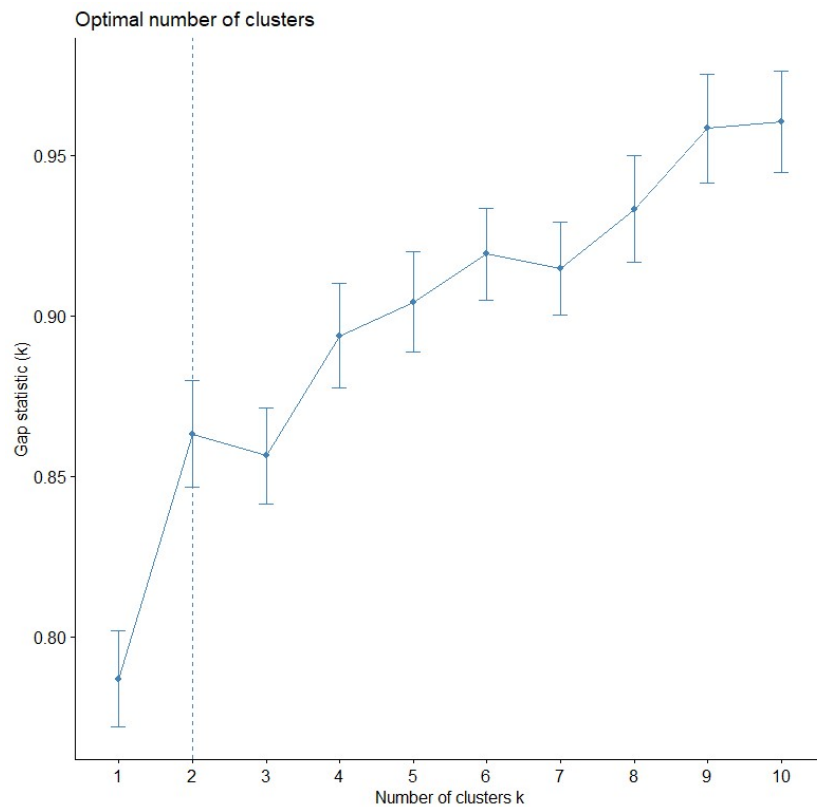
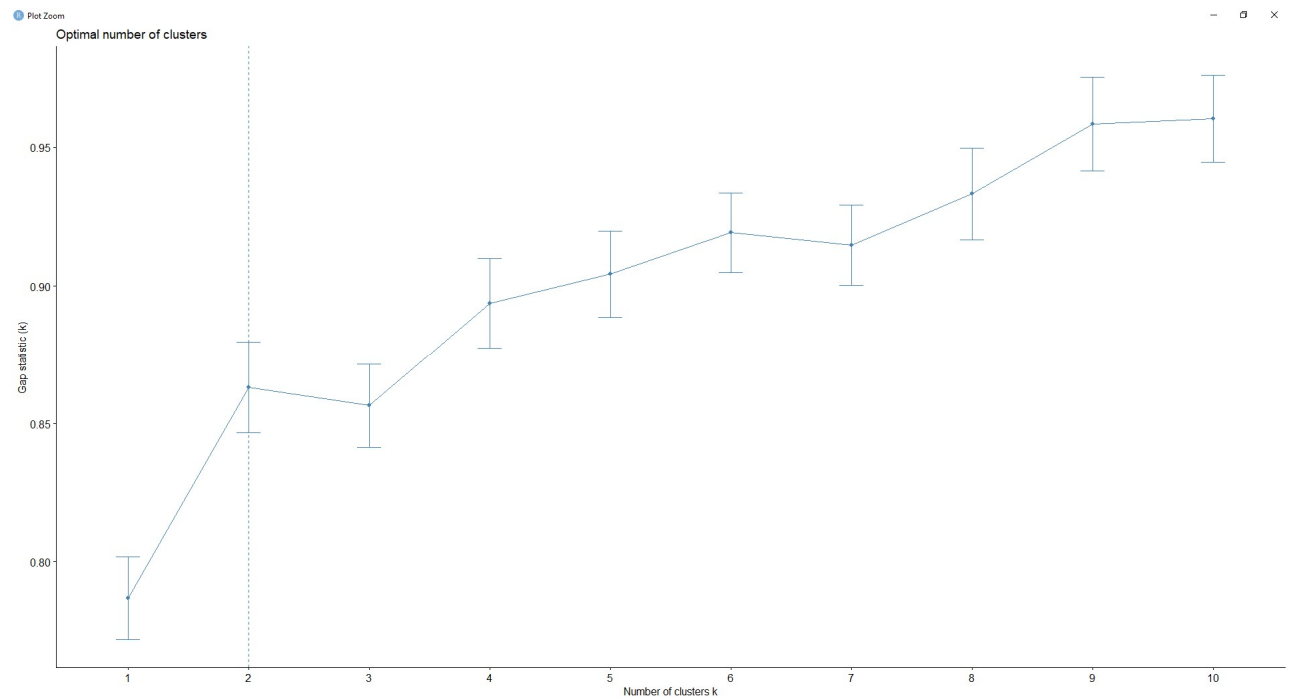
`fviz_nbclust(data, kmeans, method = "silhouette")`



Using Gap Statistics Method:

```
fviz_nbclust(data, kmeans, method = "gap_stat")
```

```
> fviz_nbclust(data, kmeans, method = "wss")
> fviz_nbclust(data, kmeans, method = "silhouette")
> fviz_nbclust(data, kmeans, method = "gap_stat")
Clustering k = 1,2,..., K.max (= 10): .. done
Bootstrapping, b = 1,2,..., B (= 100) [one "." per sample]:
..... 50
..... 100
```



5) Create Clusters with K-means

```
kmeans(data, centers = 2, iter.max = 100, nstart = 100)
```

```

34 kmeans(data, centers = 2, iter.max = 100, nstart = 100)
35 
36 
37 [Top Level]

```

```

R 4.2.2 - D:\Semester\10\ Spring 2023\Introduction To Data Science\Final Term Project / #
[124.] -0.72378317 0.1730301047 -1.022340106 -0.53370320 9.334713074 -0.073034204 1.003730413 -0.373333403
[125.] -0.66379294 -0.3104105103 -0.449858655 -0.33417399 0.033992586 -0.402227929 0.353564668 -0.897028289
[ reached getOption("max.print") -- omitted 42 rows ]
attr(,"scaled:center")
   child_mort    exports      health     imports      income    inflation  life_expect total_fer
   38.270060    41.108976    6.815689    46.890215  17144.688623    7.781832   70.555689    2.947964
attr(,"scaled:scale")
   child_mort    exports      health     imports      income    inflation  life_expect total_fer
   40.328931    27.412010    2.746837    24.209589  19278.067698   10.570704    8.893172   1.513848
> kmeans(data, centers = 2, iter.max = 100, nstart = 100)
K-means clustering with 2 clusters of sizes 113, 54

Cluster means:
   child_mort    exports      health     imports      income    inflation life_expect total_fer
1 -0.5724204    0.2130665    0.1008310  0.08411269  0.3286096 -0.1983743  0.5636294 -0.5711283
2  1.1978427 -0.4458614 -0.2109981 -0.17601360 -0.6876459  0.4151166 -1.1794467  1.1951389

Clustering vector:
 [1] 2 1 1 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 1 1 1 2 1 1 1 2 2 1 1 1 2 2 1 1 1 2 2 2 1 2 1 1 1 1 1 1 1 1 1 1 2 2 1 1 1 1 2 2 1 1 1
[64] 2 2 1 2 1 1 2 1 1 2 1 1 1 1 1 1 1 1 1 2 2 1 1 2 1 1 2 2 1 1 1 1 2 2 1 1 2 2 1 1 1 2 2 2 2 1 1 2 2 1 1 1 2 2 1 1 1 1 1 1 1 1
[127] 2 1 1 2 1 1 2 1 1 1 2 2 1 1 1 1 2 1 1 1 2 2 1 2 2 1 1 1 1 2 1 1 1 1 1 1 1 1 2 1 1 2 2

Within cluster sum of squares by cluster:
[1] 598.6012 316.5060
   (between_SS / total_SS =  31.1 %)

Available components:

[1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss" "betweenss"    "size"         "iter"
```

6) Create Cluster Centres

```
kr1<- kmeans(data, centers = 2, iter.max = 100, nstart = 100)
```

kr1\$centers

```

80 #Cluster Center
81 krl<- kmeans(data, centers = 2, iter.max = 100, nstart = 100)
82 krl
83 krl$centers
84

```

81:1 (Top Level) ↕

Console Terminal Background Jobs

R 4.2.2 · D:\Semester\10\ Spring 2023\Introduction To Data Science\Final Term Project\ ↗

```

Clustering vector:
[1] 2 1 1 2 1 1 1 1 1 1 1 1 1 1 1 1 1 2 1 1 1 2 1 1 1 2 2 1 2 1 1 2 2 2 1 2 1 1 1 1 1 1 1 1 2 2 1 1 1 2 2 1 1 1 2 1 1 1
[64] 2 2 1 2 1 1 2 1 1 2 1 1 1 1 1 1 2 2 1 1 2 1 1 2 2 1 1 1 1 2 2 1 1 2 1 2 1 1 1 2 1 1 2 2 2 1 1 2 2 1 1 1 1 1 1 1 1 1
[127] 2 1 1 2 1 1 2 1 1 1 2 2 1 1 1 1 1 2 2 1 1 1 1 2 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1

```

within cluster sum of squares by cluster:

```

[1] 598.6012 316.5060
(between_SS / total_SS = 31.1 %)

```

Available components:

```

[1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss" "betweenss"    "size"         "iter"
[9] "ifault"
> krl$centers
  child_mort    exports    health    imports    income    inflation life_expect total_fer
1 -0.5724204  0.2130665  0.1008310  0.08411269  0.3286096 -0.1983743  0.5636294 -0.5711283
2  1.1978427 -0.4458614 -0.2109981 -0.17601360 -0.6876459  0.4151166 -1.1794467  1.1951389
>

```


7) Create Hierarchical Clustering

```
d <- dist(data, method = "euclidean")
```

d

```
> d <- dist(data, method = "euclidean")
> d
```

	1	2	3	4	5	6
2	4.1264121					
3	3.8800140	1.8361116				
4	2.9862380	4.9990771	3.8653037			
5	4.3540860	1.0305446	2.0672598	4.9427724		
6	4.1354402	2.2562959	1.8652932	4.8942816	2.8438904	
7	3.9908261	0.9697675	1.5110182	4.6603515	1.5901960	2.3458549

```
hc1 <- hclust(d, method = "complete")
```

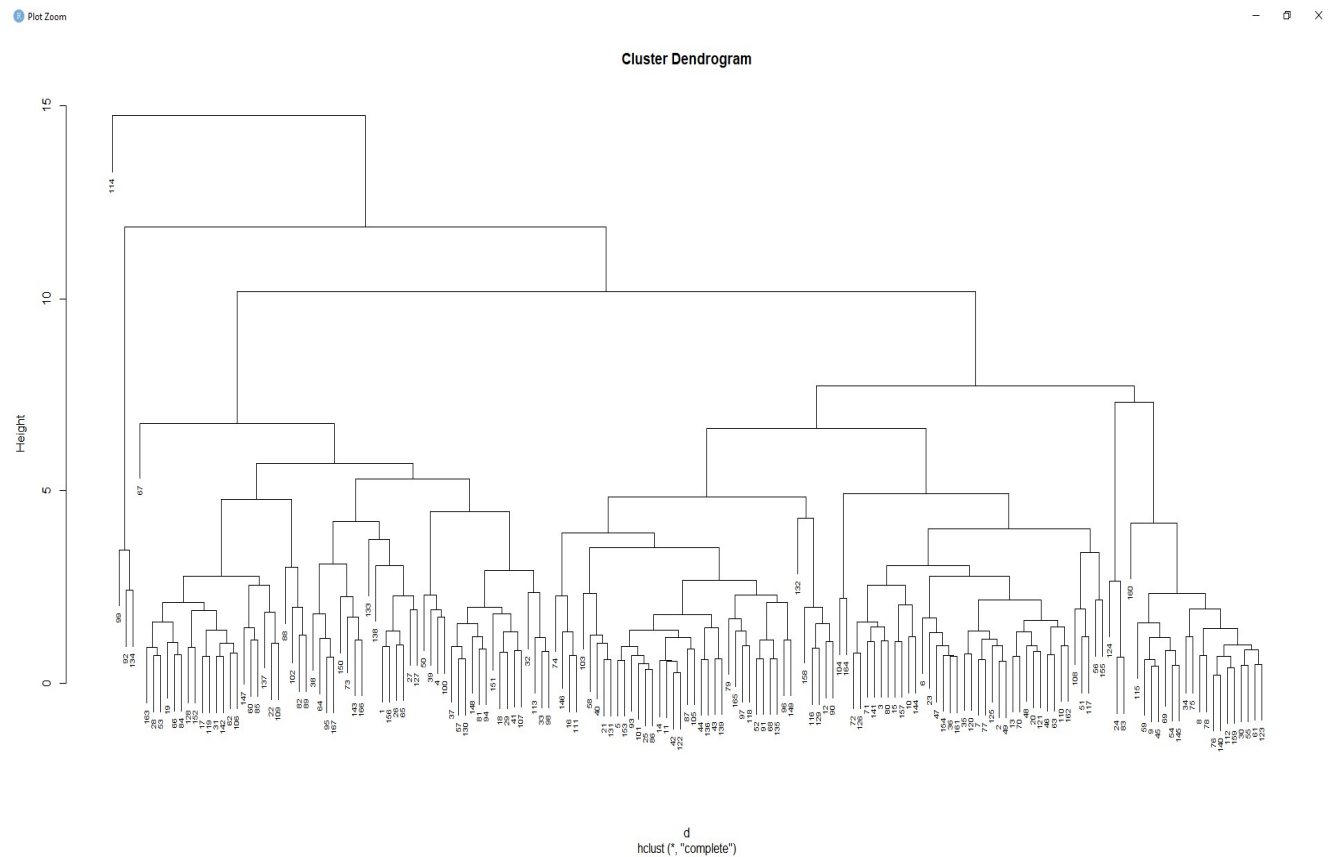
hc1

```
> hc1 <- hclust(d, method = "complete")
> hc1
```

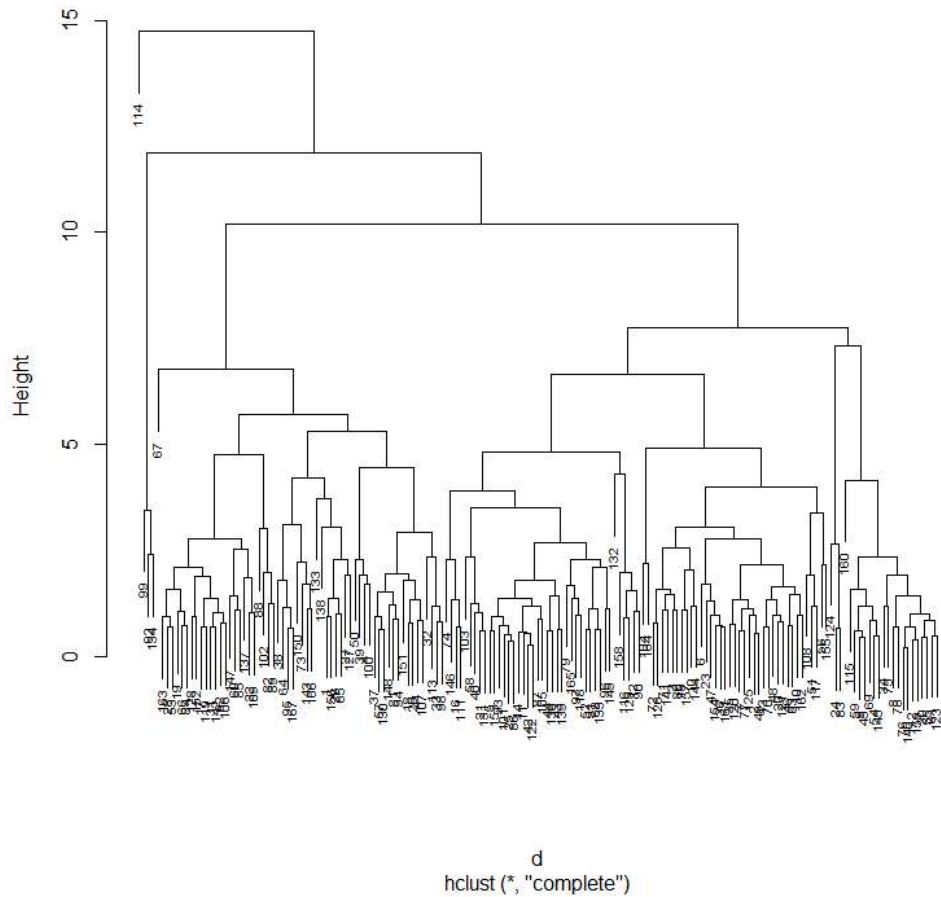
Call:
hclust(d = d, method = "complete")

Cluster method : complete
Distance : euclidean
Number of objects: 167

```
plot(hc1, cex = 0.6)
```



Cluster Dendrogram



8) Create the Clustering Result

```
kR<- pam(data,k=2)
```

```
kR
```

```
57 kR<- pam(data,k=2)
58 kR
59 4
57:1 (Top Level) 5
```

```
R 4.2.2 - D:/Semester/10/ Spring 2023/Introduction To Data Science/Final Term Project/
> kR<- pam(data,k=2)
> kR
Medoids:
  ID child_mort  exports  health  imports  income  inflation life_expec  total_fer
[1,] 60  0.9033202 -0.42349963 -0.5809185 -0.04090177 -0.7306069  0.8342082 -0.9395623  0.8732953
[2,] 122 -0.8001715 -0.03680781  0.2345648 -0.19786437  0.2414823 -0.5791319  0.6459238 -1.0159306
Clustering vector:
[1] 1 2 2 1 2 2 2 2 2 2 2 2 2 2 2 2 1 2 1 2 2 2 2 1 1 1 1 2 2 2 1 1 2 2 2 2 1 1 2 2 2 2 2 2 1 2 1 1 2 2 2 2 1 2 1 1 2 2 2 2 1 1 2 2 2 2 2
[64] 1 1 1 1 2 2 1 1 2 2 2 2 2 2 1 1 1 2 1 1 2 2 1 1 2 2 2 2 1 1 2 2 2 2 1 2 2 1 1 1 2 2 1 1 2 2 1 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
[127] 1 1 2 1 2 2 1 2 2 2 1 1 2 2 1 2 1 2 2 2 1 1 2 2 1 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
objective function:
  build  swap
2.180968 2.106798
Available components:
[1] "medoids" "id.med" "clustering" "objective" "isolation" "clusinfo" "silinfo" "diss" "call"
[10] "data"
```

summary(kR)

```

59 summary(kr)
60
59:1 (Top Level)
Console Terminal Background Jobs
R 4.2.2 · D:\Semester\10\ Spring 2023\Introduction To Data Science\Final Term Project
> summary(kr)
Medoids:
      ID child_mort    exports    health    imports    income    inflation life_expec    total_fer
[1,] 60  0.9033202 -0.42349963 -0.5809185 -0.04090177 -0.7306069  0.8342082 -0.9395623  0.8732953
[2,] 122 -0.8001715 -0.03680781  0.2345648 -0.19786437  0.2414823  -0.5791319  0.6459238 -1.0159306
Clustering vector:
 [1] 1 2 2 1 2 2 2 2 2 2 2 1 2 2 2 2 1 2 2 1 2 2 2 1 1 1 1 2 2 1 1 2 2 2 1 1 1 2 1 2 2 2 2 2 1 2 1 1 2 2 2 1 1 2 2 2 2 2
[64] 1 1 1 1 2 2 1 1 2 1 2 2 2 2 2 2 1 1 2 2 1 1 1 2 2 1 1 2 2 1 2 2 2 1 2 2 1 2 2 2 1 2 2 1 1 1 2 2 1 1 1 2 2 1 1 2 2 2 2 2
[127] 1 1 2 1 2 2 1 2 2 2 1 1 2 2 1 2 2 2 1 1 2 1 1 1 2 2 1 1 2 2 2 2 1 1 1 2 2 1 1 1 2 2 1 1 2 2 2 1 1 2 2 2 2 2
Objective function:
      build      swap
2.180968 2.106798

Numerical information per cluster:
      size max_diss av_diss diameter separation
[1,] 69 8.531029 2.268592 10.58830 0.7034727
[2,] 98 8.527658 1.992882 10.82467 0.7034727

Isolated clusters:
L-clusters: character(0)
L*-clusters: character(0)

Silhouette plot information:
      cluster neighbor    sil_width
64      1      2 0.465337895
107     1      2 0.463982247
95      1      2 0.460753568
26      1      2 0.449869300
167     1      2 0.444260686
1       1      2 0.439117097
148     1      2 0.438239046
29      1      2 0.432664327
98      1      2 0.428458744
41      1      2 0.425128224

Average silhouette width per cluster:
[1] 0.2395250 0.3416712
Average silhouette width of total data set:
[1] 0.2994671

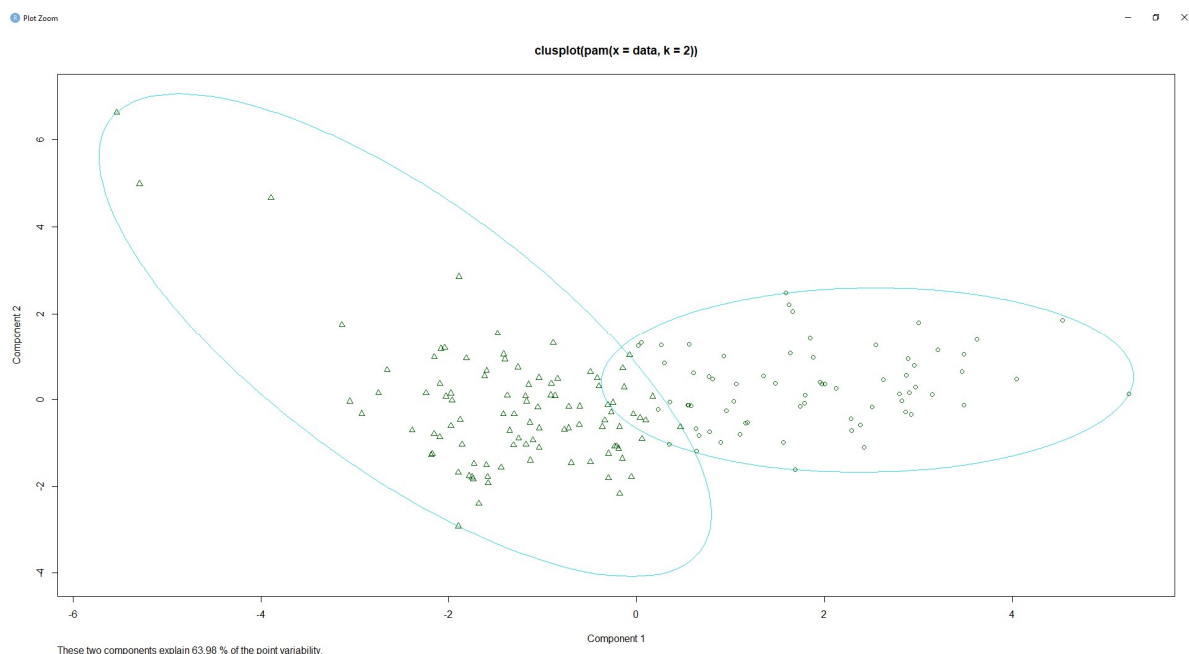
Available components:
[1] "medoids" "id.med" "clustering" "objective" "isolation" "clusinfo" "silinfo" "diss" "call"
[10] "data"

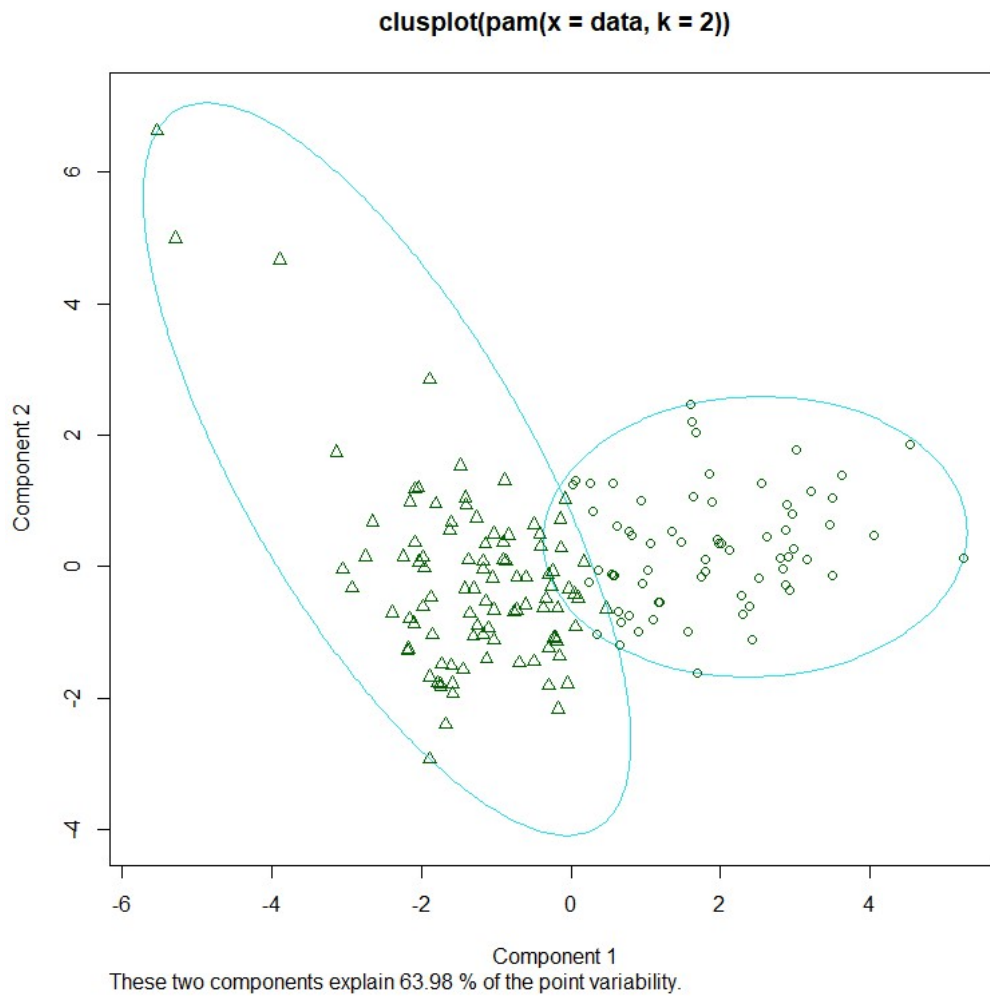
```

10) Generate the Cluster and Silhouette Plot

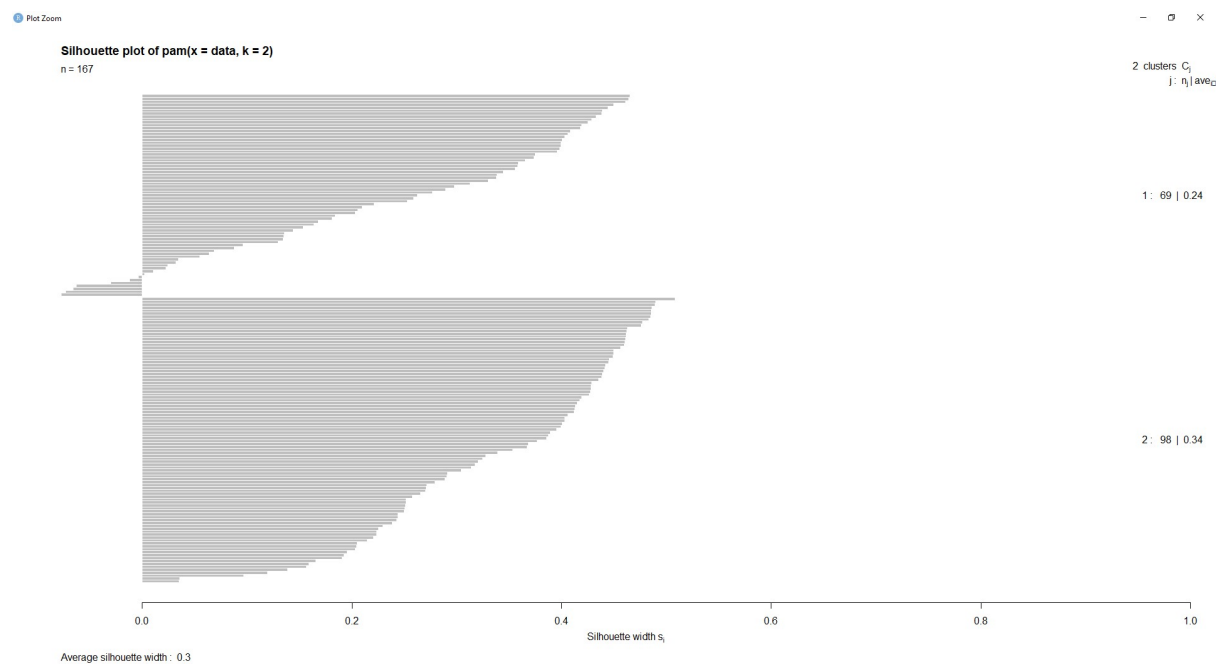
$$\text{plot(kR)}$$

Cluster Plot:





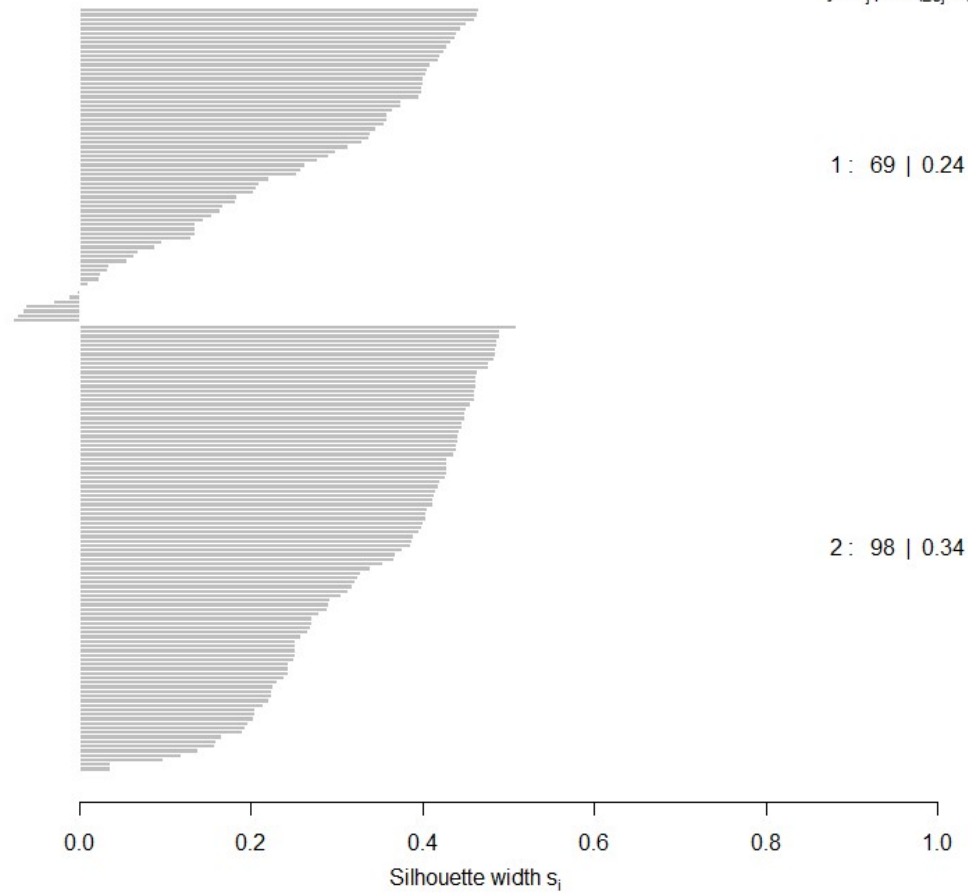
Silhouette Plot:



Silhouette plot of pam(x = data, k = 2)

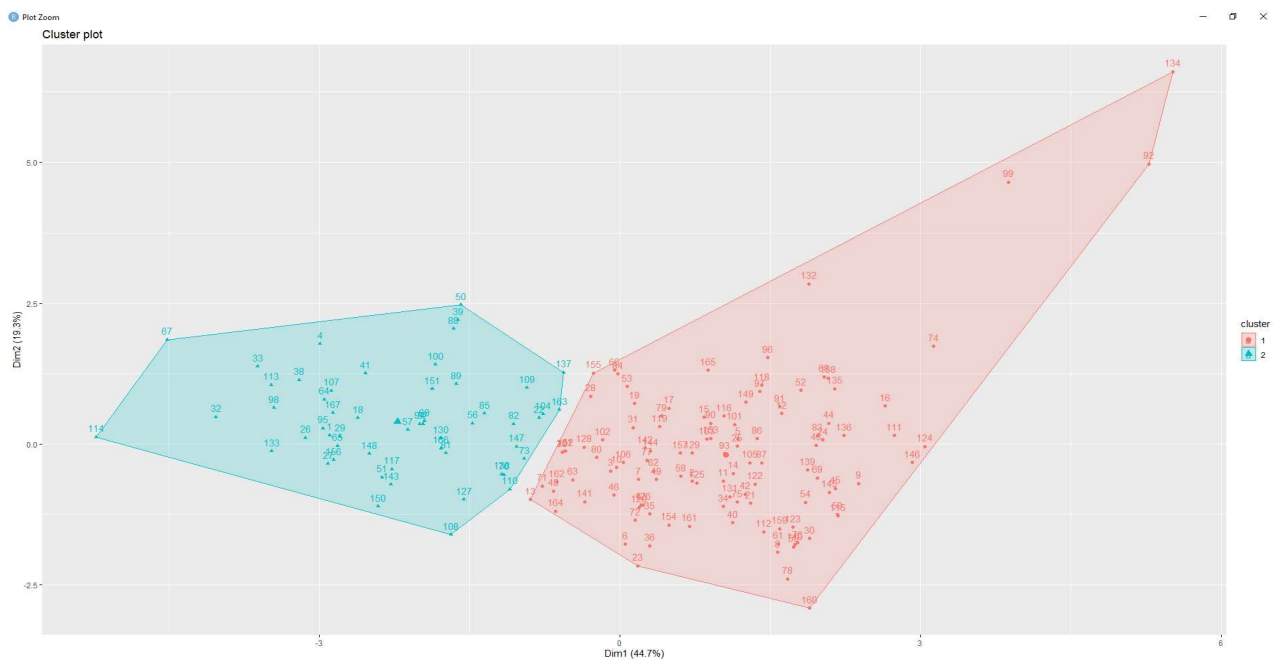
n = 167

2 clusters C_j
 $j: n_j | \text{ave}_{i \in C_j} s_i$



11) Generate the Cluster Biplot

`fviz_cluster(kmeans(data, centers = 2, iter.max = 100, nstart = 100), data=data)`




```
Dataset1 <- Dataset |> mutate(cluster = clusters$cluster)
```

Dataset1

53Dataset1 <- Dataset |> mutate(cluster = clusters\$cluster)

54Dataset1

55

53:1 (Top Level) ▾

ConsoleTerminalBackground Jobs

R 4.2.2 · D:/Semester/10) Spring 2023/Introduction To Data Science/Final Term Project/

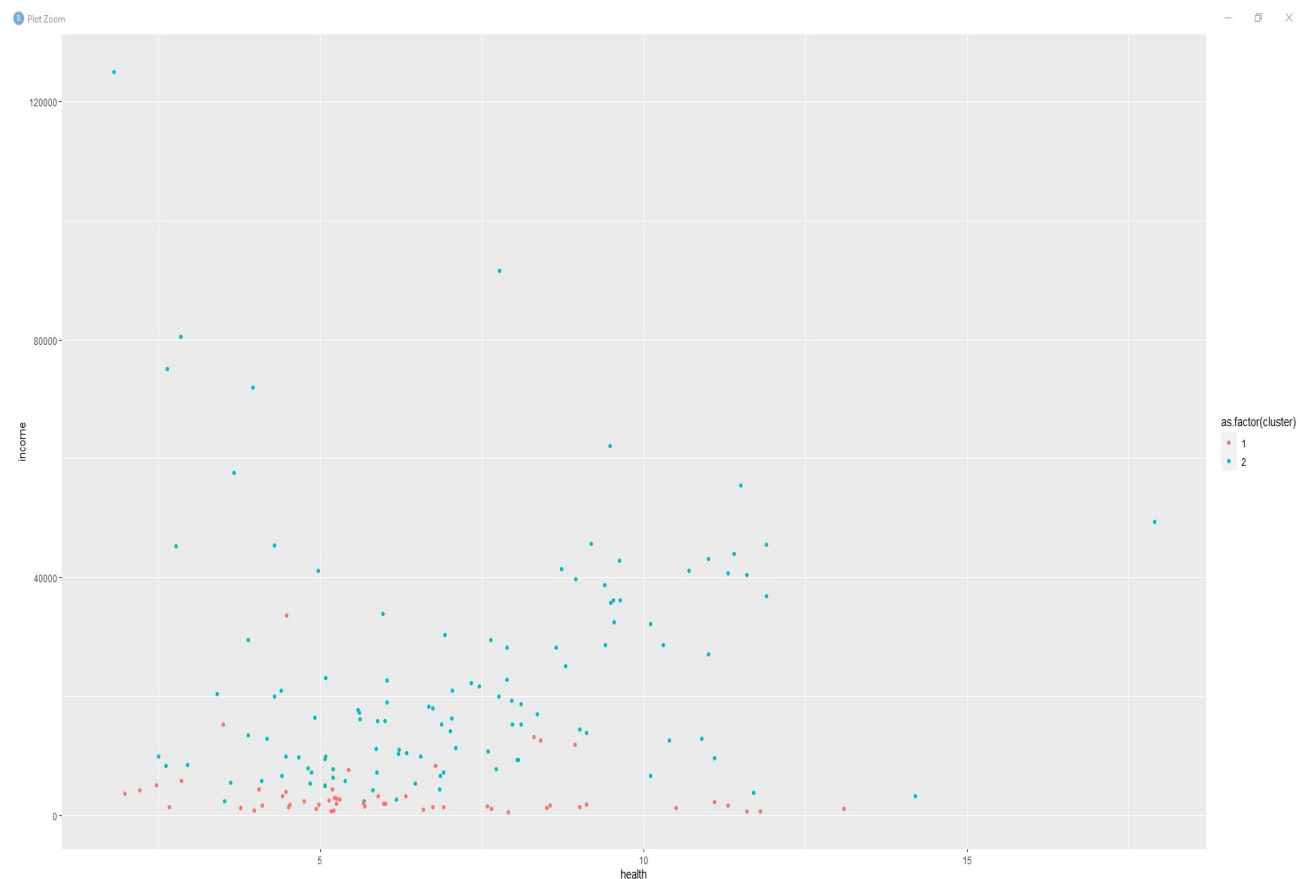
> Dataset1 <- Dataset |> mutate(cluster = clusters\$cluster)

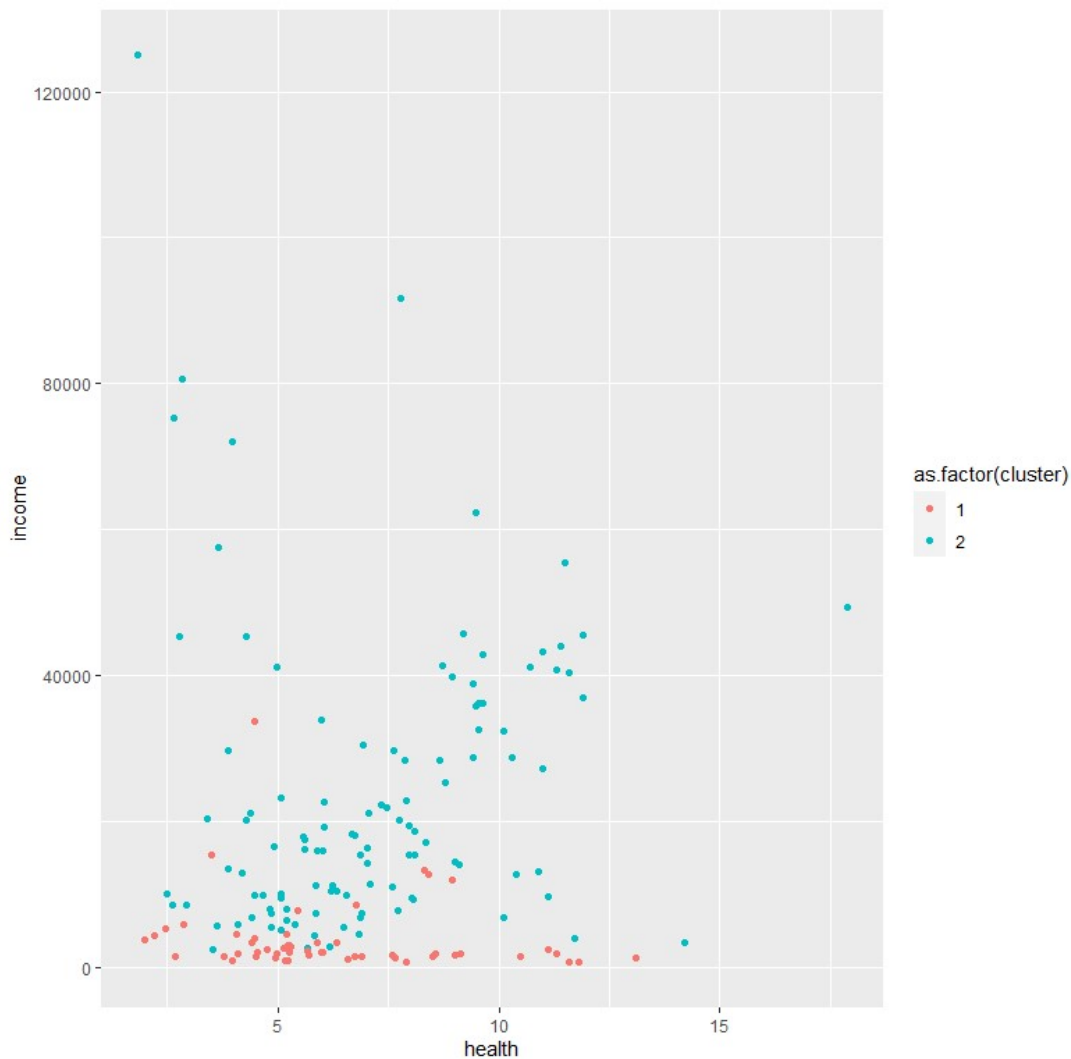
> Dataset1

	country	child_mort	exports	health	imports	income	inflation	life_expec	total_fer	gdpp	cluster
1	Afghanistan	90.2	10.00	7.58	44.9	1610	9.440	56.2	5.82	553	1
2	Albania	16.6	28.00	6.55	48.6	9930	4.490	76.3	1.65	4090	2
3	Algeria	27.3	38.40	4.17	31.4	12900	16.100	76.5	2.89	4460	2
4	Angola	119.0	62.30	2.85	42.9	5900	22.400	60.1	6.16	3530	1
5	Antigua and Barbuda	10.3	45.50	6.03	58.9	19100	1.440	76.8	2.13	12200	2
6	Argentina	14.5	18.90	8.10	16.0	18700	20.900	75.8	2.37	10300	2
7	Armenia	18.1	20.80	4.40	45.3	6700	7.770	73.3	1.69	3220	2
8	Australia	4.8	19.80	8.73	20.9	41400	1.160	82.0	1.93	51900	2
9	Austria	4.3	51.30	11.00	47.8	43200	0.873	80.5	1.44	46900	2
10	Azerbaijan	39.2	54.30	5.88	20.7	16000	13.800	69.1	1.92	5840	2
11	Bahamas	13.8	35.00	7.89	43.7	22900	-0.393	73.8	1.86	28000	2
12	Bahrain	8.6	69.50	4.97	50.9	41100	7.440	76.0	2.16	20700	2
13	Bangladesh	49.4	16.00	3.52	21.8	2440	7.140	70.4	2.33	758	2
14	Barbados	14.2	39.50	7.97	48.7	15300	0.321	76.7	1.78	16000	2
15	Belarus	5.5	51.40	5.61	64.5	16200	15.100	70.4	1.49	6030	2
16	Belgium	4.5	76.40	10.70	74.7	41100	1.880	80.0	1.86	44400	2

13) Generate the Cluster Visualization

```
Dataset1 |> ggplot(aes(x = health, y = income, col = as.factor(cluster))) + geom_point()
```





Conclusion: The most well-liked clustering technique is probably K-Means. Studying its features is therefore important for the categorization, data mining, and machine learning communities as well as the growing number of practitioners in marketing research, bioinformatics, customer management, engineering, and other application domains. As with other unsupervised learning, the proper number of clusters is one of the most contentious clustering concerns, which some may consider as being unfounded since, in many circumstances, "clusters are not in data but in the viewing eye." Even when there is little information available about the data, it is the algorithm that can group data points into categories the fastest and most effectively. In turn, taking into account the proper algorithm may help to achieve results that are more accurate while also saving time and work.

References:

- [1] <https://www.kaggle.com/datasets/rohan0301/unsupervised-learning-on-country-data>
- [2] Likas, A., Vlassis, N. and Verbeek, J.J., 2003. The global k-means clustering algorithm. Pattern recognition, 36(2), pp.451-461.
- [3] Dundar, A., Jin, J. and Culurciello, E., 2015. Convolutional clustering for unsupervised learning. arXiv preprint arXiv:1511.06241.
- [4] Bradley, P.S., Bennett, K.P. and Demiriz, A., 2000. Constrained k-means clustering. Microsoft Research, Redmond, 20(0), p.0.